

Yufei Wu

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Education

University of Edinburgh

M.Sc. Computer Science

Edinburgh, UK

09/2025-present

Courses: Current: Machine Learning Practical, Accelerated Natural Language Processing, Distributed Systems, Computer Graphics: Rendering, Human-Computer Interaction; Next semester: Algorithmic Game Theory and its Applications, Machine Learning Systems

University of Auckland

B.S. Computer Science

Auckland, NZ

07/2022-09/2025

Double major in Statistics

07/2022-09/2025

7.542/9.0 GPA

Core courses: Machine Learning(A-) Discrete Structures in Mathematics and Computer Science(A) Data Analysis(A+) Statistical Computing(A+) Statistical Modeling(A+)

UP International College

The UOA Certificate in Foundation Studies(Level 3)

Auckland, NZ

07/2021-06/2022

Projects

Faultless Document Reviewer

University of Auckland

06/2025

- Developed an intelligent document analysis system integrating Google Gemini 2.0, LangChain, and ChromaDB to automate engineering report quality assurance.
- Implemented Retrieval-Augmented Generation (RAG) pipeline using ChromaDB's HNSW indexing to process 50-page documents in <10 seconds, reducing API costs through semantic chunking and targeted context.
- Designed 5-stage filtering pipeline (validation, redundancy removal, confidence thresholding, trivial filtering, priority ranking) to reduce AI hallucinations.
- Built dual interface (CLI and Word add-in) with pluggable storage backends(CSV/SQLite).

Distributed Fault-Tolerant Prime-Finding System

University of Edinburgh

11/2025

- Built an AFS user-space distributed file system backed by a Raft cluster to provide durable, linearizable open/create/read/write/close RPCs, plus whole-file client caching with version-based TestAuth validation.
- Implemented a fault-tolerant RPC layer with timeouts, retry/backoff, round-robin replica selection, and idempotent ops via unique request IDs.
- Implemented a coordinator-worker prime-finding pipeline over TCP
- Added Chandy-Lamport snapshots and persisted state in the DFS for fast recovery.

Algorithmic Trading Bot

Hack The Burgh 2025 (Optiver Challenge)

11/2025

- Built Python-based algorithmic trading bot integrating Hugging Face Transformers, Optibook API, and custom risk management to compete in 20-hour live trading simulation.
- Designed three-strategy system: (1) market making with dynamic spread adjustment, (2) NLP-driven sentiment trading using zero-shot classification, (3) statistical arbitrage exploiting dual-listing price inefficiencies.
- Built performance tracking system with automated CSV exports, real-time PnL visualization, and statistical analysis.

Experience

HBISDT, XINDA(HANDAN) Technology Intelligent Manufacturing R&D Team

Handan, CHINA

Research Internship

01/2024-01/2024

- Designed and implemented an interface with TensorFlow to facilitate the import of steel material images. Categorised data into training and prediction sets, extracted features from training set images, and conducted comprehensive analysis. Implemented a CNN(Convolutional Neural Network) classification model to automatically assess the internal low-magnification quality grades of rolled plates, determining compliance with required standards.
- Developed a foundational interface for automatically assessing the eligibility of steel coil materials. The interface allows for image importation, providing users with the flexibility to choose from ID3, C4.5, or CART algorithms for decision tree generation. It presents generated tree diagrams and result accuracies. The system conducts quality assessments of parent coils, streamlining the quality evaluation process.

Activities

Science Ambassador Programme

Auckland, NZ

Science Ambassador

03/2023-10/2023

- Part of the Co-curricular Recognition Program.
- Engaged in assisting with the 2023 Open Day event.
- Assisted new students in integrating into campus life and providing guidance to address their queries.

New Zealand Chinese Students Association

Auckland, NZ

Full stack developer

03/2024-11/2025

- Revamped official website framework.
- Assisted in enhancing backend system security.
- Contributed to the creation of the payment gateway for membership registration.

Certificates and Awards

University of Auckland, School of Science

Auckland, NZ

Certificate of Excellent Grades in Statistics, Department of Statistics

07/2024, 12/2023, 08/2023

Certificate of Excellent Grades in Mathematics, Department of Mathematics

08/2023, 01/2023

Up International College Programme

Auckland, NZ

Foundation Connect Progression Scholarship

07/2021

Skills

- Coding: Python, Java, JavaScript, C++, R
- Language skills: Mandarin(native), English(fluent)

Links

- Github: <https://github.com/char-WU>
- LinkedIn: <https://www.linkedin.com/in/yufei-charlotte-wu>